Learning-Rate Schedules Are Not Adiabatic: A Rate-Dependent Correction for Loss-Curve Prediction

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https://github.com/wjjpku/DL-final

Abstract

State-of-the-art laws for predicting the pretraining loss curve under a learning-rate (LR) schedule—the Multi-Power Law (MPL) and the Functional Scaling Law (FSL)—are *adiabatic*: they predict the quasi-static loss the schedule would reach at equilibrium. MPL does use the LR decrements (a convolution of $\Delta\eta_k$ with a saturating kernel), but its only handle on the *rate* of decay is the single empirical exponent γ; calibrated on slow schedules, this cannot capture the transient by which the loss *lags* its equilibrium when the LR is annealed fast. The result is a systematic *positive* residual on *fast* (warmup-stable-decay, WSD) decays, which we identify as a non-adiabatic relaxation lag. The classical SGD relaxation time governing this lag, $\tau \sim 1/(\eta\lambda)$, is well known; our contribution is to (i) show it governs the residual of adiabatic loss-curve laws, (ii) derive the resulting correction in closed form from AdamW's preconditioned second-moment dynamics—a convolution of the per-step LR decrements with a cumulative-LR (S-time) exponential kernel, $\mathrm{DropRelaxS}$, whose amplitude equals the exactly-identified equilibrium-floor sensitivity $dL_{\mathrm{eq}}/d\eta$—and (iii) turn it into a usable predictor. We confirm in LLM pretraining that the residual relaxes with the predicted $\tau \propto 1/\eta$ (measured pooled exponent $p = 1.00 \pm 0.18$), which also self-consistently establishes the noise-dominated regime the derivation assumes. Because the kernel's rate constant is a *scale-invariant preconditioned curvature*, the correction can be *predicted, not fitted*, and transferred across model size: with zero fitting on the target curve it improves held-out sharp-decay loss prediction by 28–56% (44% on average). The correction also explains MPL's one unexplained empirical exponent, γ, as its implicit low-capacity surrogate. We are candid about scope: the gain is in the regime where WSD curves are unavailable; given them, a refit MPL absorbs the residual.

1 Introduction

The learning-rate schedule shapes the *entire* loss trajectory of LLM pretraining [11, 12, 13]. *Schedule-aware loss-curve prediction* asks: given $\{\eta_t\}$, predict $L(t)$, so that schedules can be compared and optimized *before* spending compute. The defining test is cross-schedule generalization—fit on cosine schedules, predict the unseen WSD family. The state of the art is the **Multi-Power Law** (MPL) [2], with a matching kernel-regression theory, the **Functional Scaling Law** (FSL) [3]:

$$L(t) = L_0 + A\,S(t)^{-\alpha} + B\sum_{k\le t}\Delta\eta_k\,G\big(\eta_k^{-\gamma}(S(t)-S(k))\big), \quad G(x) = 1-(1+Cx)^{-\beta}, \tag{1}$$

where $S(t) = \sum_{i\le t}\eta_i$ and $\Delta\eta_k = \eta_{k-1} - \eta_k$.

The gap: existing laws are adiabatic. These laws *do* use the LR decrements: MPL convolves $\Delta\eta_k$ with a saturating kernel G, and its factor $\eta_k^{-\gamma}$ is a rate-of-decay knob. But this single empirical

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exponent is their *only* handle on how *fast* the LR changes. Structurally they predict the *quasi-static* loss—the value the schedule would reach if the loss stayed at equilibrium; the river-valley analysis [4] makes this explicit, modelling the noise penalty as set *instantaneously* by the current LR. We call such laws *adiabatic*. Calibrated on slow schedules, γ does not extrapolate to fast WSD decays, where the loss genuinely lags equilibrium—leaving a structured residual (Fig. 1) that no single rate exponent captures across schedules. Our correction is the derived transient that γ only crudely surrogates.

Contributions.

- We show that adiabatic loss-curve laws leave a systematic *positive* residual on fast decays, and identify it as a non-adiabatic relaxation lag (§2).
- From AdamW's preconditioned second-moment dynamics we *derive* the leading non-adiabatic correction in closed form, the rate-dependent term $\mathrm{DropRelaxS}$, with amplitude equal to the exactly-identified $dL_{\mathrm{eq}}/d\eta$ (§3). The relaxation time it uses is the classical $\tau \sim 1/(\eta\lambda)$; our point is that it governs the *loss-curve-law residual*, not that the timescale is new.
- We confirm in LLM pretraining that the residual relaxes with $\tau \propto 1/\eta$ ($p = 1.00 \pm 0.18$), self-consistently establishing its noise-dominated regime (§4).
- The kernel's rate constant is a scale-invariant preconditioned curvature, enabling a *parameter-free, cross-scale* prediction that improves held-out sharp-decay loss by 44% with no target-curve fitting (§4).
- The correction explains MPL's empirical γ as its implicit surrogate, and we give a fair-baseline diagnostic separating a genuine formula gain from an information gap (§4,5).

All experiments use the public MPL loss curves (Llama-2-style $25/100/400$M, 9 schedules each) [2]; a faithfully reproduced MPL (test MAE $0.0054/0.0050/0.0085$, matching [2]) is the baseline throughout.

2 The non-adiabatic residual

We take a *well-fit* MPL (the published parameters [2]) and examine its prediction residual $r(t) = L_{\mathrm{true}}(t) - L_{\mathrm{MPL}}(t)$ on held-out schedules. (Isolating the non-adiabatic term requires a well-fit MPL: a baseline that already captures the *adiabatic* part cleanly, so the remainder is the lag and not a backbone mis-fit; a cosine-only fit conflates the two and gives a far noisier residual.) The residual has a sharp signature (Fig. 1): it is ≈ 0 throughout the long stable phase and on gradual cosine decay, but jumps *positive* exactly when the LR decays steeply, and is largest—and *undamped at the curve's end*—for the sharpest decays (`wsd`, `wsdld`). The end-of-curve lag is $+(6\text{–}15) \times 10^{-3}$ on WSD/WSDLD versus $\lesssim 4 \times 10^{-3}$ on cosine. Crucially, a gradual cosine and a sharp WSD decay reach the *same* final LR, yet only the fast one lags: same destination, faster sweep, bigger lag—the textbook signature of a rate-dependent (non-adiabatic) effect, and exactly the dependence an adiabatic law cannot represent.

3 Theory: the leading non-adiabatic correction

MPL is the adiabatic limit. At a fixed LR the loss relaxes toward a quasi-static value $L_{\mathrm{eq}}(\eta)$ (backbone progress plus the equilibrium variance of the sharp "mountain" directions of the river-valley landscape [4]). On schedules that vary the LR slowly the loss tracks L_{eq}, so **a well-fit MPL approximates the adiabatic** L_{eq} (its kernel, in particular γ, is calibrated to reproduce the quasi-static component). The residual on *fast* held-out decays is the non-adiabatic remainder.

Derivation (AdamW). In the Hessian eigenbasis, mode i has curvature λ_i and gradient-noise variance s_i^2. With $\beta_2 \to 1$ the Adam preconditioner sits at $v_i^* = \mathbb{E}[g_i^2] = \lambda_i^2 V_i + s_i^2$, and in the noise-dominated mountain directions $v_i^* \approx s_i^2$, so Adam reduces to SGD with a *preconditioned* step $\eta_i^{\mathrm{eff}} = \eta/s_i$ [7, 9]. The second moment $V_i = \mathbb{E}[\delta_i^2]$ then obeys the *exact* recursion $V_i(t{+}1) = (1 - \eta_i^{\mathrm{eff}}\lambda_i)^2 V_i(t) + (\eta_i^{\mathrm{eff}})^2 s_i^2$, with fixed point $V_i^*(\eta)$ and the classical relaxation rate $2\eta_i^{\mathrm{eff}}\lambda_i$, i.e. $\tau_i \sim 1/(\eta\,\lambda_i/s_i)$ [6]. Writing the deviation $\Delta_i = V_i - V_i^*(\eta_t)$ and using $(1 - \eta^{\mathrm{eff}}\lambda)^2 \approx e^{-2\eta^{\mathrm{eff}}\lambda}$

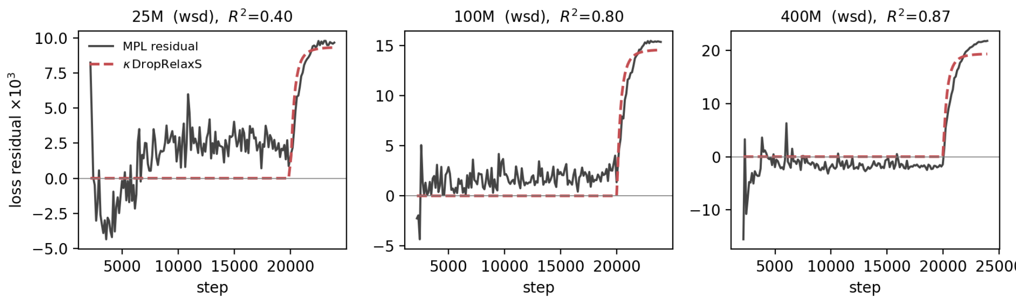

Figure 1: The well-fit MPL prediction residual on the held-out `wsd` curve (solid) is ≈ 0 during the stable phase and jumps up at the step-20000 decay. The derived rate-dependent term $\kappa\,\mathrm{DropRelaxS}$ (dashed) captures it (R^2=$0.40/0.80/0.87$ at $25/100/400$M; the signal, and the fit, grow with model size).

(valid because the lag lives on *slow* modes, $\eta\lambda_i/s_i \ll 1$, verified post hoc), the linear recursion solves to a cumulative-LR (S-time) exponential:

$$\Delta_i(t) = \frac{dV_i^*}{d\eta} \sum_{t'} e^{-2(\lambda_i/s_i)(S_t - S_{t'})}\, \mathrm{drop}_{t'}, \qquad \mathrm{drop}_{t'} = (\eta_{t'-1} - \eta_{t'})_+. \tag{2}$$

The loss lag $\Delta L = \frac{1}{2}\sum_i \lambda_i \Delta_i$ is therefore a w_i-weighted *mixture* of S-time exponentials with rates $2\lambda_i/s_i$, and the total amplitude is an *exact identity*:

$$\Delta L(t) = \sum_i w_i \big[e^{-2(\lambda_i/s_i)(S-S')} \circledast \mathrm{drop}\big](t), \qquad \sum_i w_i = \frac{dL_{\mathrm{eq}}}{d\eta}. \tag{3}$$

Collapsing the spectrum to one effective rate gives the operational law (one shape constant λ_{slow}, one calibration $c \leq 1$):

$$\boxed{L(t) = L_{\mathrm{MPL}}(t) + c \cdot \eta_{\mathrm{peak}} \frac{dL_{\mathrm{eq}}}{d\eta} \cdot \underbrace{\sum_{t'} e^{-\lambda_{\mathrm{slow}}(S_t - S_{t'})}\, \mathrm{drop}_{t'}}_{\mathrm{DropRelaxS}_{\lambda_{\mathrm{slow}}}(t)}} \tag{4}$$

What the derivation predicts (all tested in §4). (i) *Sign*: $dV_i^*/d\eta > 0$, so $\Delta L > 0$ (loss *above* MPL). (ii) *Rate*: the per-step rate is $2(\eta/s_i)\lambda_i \propto \eta$, hence $\tau \propto 1/\eta$; a self-consistent treatment of the preconditioner gives $\tau \propto \eta^{-p}$ with p=1 for noise-dominated modes and p=$2/3$ if signal-dominated. (iii) *Amplitude*: $\sum w_i = dL_{\mathrm{eq}}/d\eta$, measurable from the noise floor. (iv) *Shape*: $\lambda_{\mathrm{slow}} = 2(\lambda/s)_{\mathrm{eff}}$ is a *preconditioned* curvature—the quantity Adam stabilizes [9]—so it is roughly scale/architecture invariant. (v) *Spectral mixture*: Eq. (3) is a sum of exponentials, so a two-exponential kernel should beat one.

4 Experiments

We freeze the published MPL and compare it to MPL+Eq. (4); the only addition is the rate-dependent term. Unless noted λ_{slow}=10.

(i) The rate prediction $\tau \propto 1/\eta$ holds (Fig. 2). The two-stage `wsdcon` curves step the LR down at step 8000 to a constant $\eta \in \{3, 9, 18\} \times 10^{-5}$, giving a clean relaxation window. Fitting the residual transient $r(t) \propto e^{-(t-8000)/\tau}$ yields τ that scales as η^{-p} with *pooled* $p = 1.00 \pm 0.18$ (per scale $-1.06/-0.94/-0.51$ at $100/400/25$M). This is the classical $\tau \sim 1/(\eta\lambda)$ confirmed for the loss-curve-law residual in LLM pretraining, and—since p=1 is the noise-dominated value—it self-consistently justifies the $v_i^* \approx s_i^2$ step of the derivation. The implied rate constant $\lambda_{\mathrm{slow}} = 1/(\tau\eta) \approx 10$ is scale-invariant; its modes have $\eta_{\mathrm{peak}}\lambda_{\mathrm{eff}} \approx 2\times10^{-3} \ll 2$ (deep in the slow regime, far from the edge of stability—unlike γ).

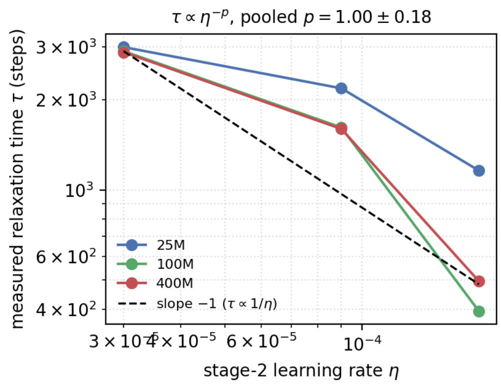

Figure 2: Loss-relaxation time τ measured from `wsdcon` decay transients vs. stage-2 LR. The 100/400M curves track the predicted slope -1 ($\tau \propto 1/\eta$); pooled $p = 1.00 \pm 0.18$.

Table 1: Parameter-free cross-scale prediction: well-fit MPL vs. MPL+Eq. (4) with *no* fitting on the target curve. MAE on held-out sharp decays.

	`wsd`		`wsdld`		
Scale	MPL	+ours	MPL	+ours	avg. Δ
25M	0.00341	0.00246	0.00314	0.00219	-29%
100M	0.00375	0.00164	0.00325	0.00161	-53%
400M	0.00470	0.00236	0.00385	0.00212	-47%
				overall	$-\mathbf{44}\%$ (6/6)

(i′) The same scaling holds in a from-scratch AdamW simulation. To check that $\tau \propto 1/\eta$ is the mechanism and not a data artefact, we simulate AdamW on a noisy quadratic, equilibrate at a high LR, step it down, and fit the loss-relaxation time (`repro/beta2_ablation.py`). Sweeping the post-step LR reproduces $\tau \propto 1/\eta$ with log-log slope $p = 1.01$. The same simulation also *rules out* a β_2 relaxation channel: sweeping $\beta_2 \in [0.9, 0.9999]$ leaves τ flat (CV 2%), because the lag lives on noise-dominated modes where the Adam preconditioner $v \approx s^2$ is constant. (The β_2 sensitivity of cooldown reported by Dremov et al. [5] is an *outcome*/bias–variance effect, distinct from the relaxation time.) The two-exponential improvement in (ii) is therefore the curvature-spectral spread, not β_2.

(ii) The residual is the derived term. Regressing the well-fit MPL residual on $\mathrm{DropRelaxS}_{10}$ (through the origin) gives $R^2{=}0.40/0.80/0.87$ at $25/100/400$M (Fig. 1); a 2-exponential kernel improves R^2 by 0.06–0.07, confirming the predicted spectral mixture (Eq. 3).

(iii) The amplitude is predictable from the noise floor. Estimating $dL_{\mathrm{eq}}/d\eta$ *independently* from the two-stage final losses (noise floor vs. LR) predicts the fitted κ up to a near-constant factor: the ratio $\kappa_{\mathrm{fit}}/\kappa_{\mathrm{pred}} = 0.43/0.51/0.60$ has coefficient of variation 14%, i.e. $c \approx 0.5$ is universal (the <1 is the spectral spread of Eq. 3).

(iv) Parameter-free, cross-scale prediction (Table 1). Because λ_{slow} and c are scale-invariant and $dL_{\mathrm{eq}}/d\eta$ is read off the (cheap) noise floor, the whole correction can be *predicted with zero fitting on the target curve*: take $\lambda_{\mathrm{slow}}{=}10$, c from a leave-one-scale-out average of the *other* scales, and $dL_{\mathrm{eq}}/d\eta$ from the target's own two-stage probes (distinct curves from the test). Applied to the held-out `wsd`/`wsdld` decays it cuts MAE by 28–56% (44% average, wins 6/6)—small-model constants plus a cheap probe predict the large-model sharp-decay curve $\sim 2\times$ better than MPL transfer.

(v) Honest scope: when it does *not* help. If instead one refits MPL on a few *full* WSD curves, MPL itself absorbs the residual (held-out MAE $0.0037 \to 0.0032$) and the explicit term adds nothing. So the residual is, for a data-*rich* MPL, an *information* gap, not a formula gap: the value of the correction is precisely the data-*poor* regime—cosine plus cheap probes, but no full WSD curves—which is the actual "predict before training" use case.

(vi) Relation to γ. MPL's empirical factor $\eta_k^{-\gamma}$ gives its kernel a weak dependence on the LR at a decrement—an *implicit, low-capacity* handle on exactly this non-adiabatic lag. This explains both why γ is essential (3–4$\times$ worse without it) yet underivable from the (adiabatic) spectral theory, and why a data-rich MPL can absorb the residual through γ while a few-shot one cannot.

5 Related work

Loss-curve laws. Tissue et al. [1], MPL [2], and FSL [3] predict the trajectory from integrated-LR statistics (cumulative LR / intrinsic time / convolutional functionals). FSL and its follow-ups also derive optimal schedules [3]. All are adiabatic in our sense—functions of integrals of $\{\eta_t\}$—and our rate-dependent term is the leading correction they omit; it also gives MPL's empirical γ a mechanism.

Cooldown dynamics and schedule geometry. The river-valley picture [4] explains qualitatively why decay "reveals progress", and explicitly models the noise penalty as *instantaneous* in the current LR—the adiabatic assumption we refine. Dremov et al. [5] study the cooldown empirically and, notably, find that raising AdamW's β_2 during cooldown helps and that the second-moment timescale interacts with the decay; this is consistent with the second, β_2-dependent relaxation channel our theory predicts (§6), though they frame it as a bias–variance trade-off rather than a non-adiabatic lag. Late-decay analyses [11] and schedule-shape searches [10] study related questions (why a long high-LR phase helps; what optimal shapes look like) but do not model the loss's rate dependence.

SGD relaxation physics. That a quadratic mode relaxes with $\tau \sim 1/(\eta\lambda)$ and that the SGD noise floor scales with η are classical, from noisy-quadratic and Langevin analyses of SGD [6]; adaptive methods inherit a preconditioned version [7], operating at an adaptive edge of stability [8, 9]. We do not claim these ingredients as new. Our contribution is the synthesis: identifying that adiabatic loss-curve laws miss exactly this relaxation lag, deriving the correction with an exactly-identified amplitude, and building a parameter-free cross-scale predictor from it.

6 Limitations

The gain is regime-specific (data-poor sharp decays); λ_{slow} and c are *measured*, not computed from first principles (this needs the noise spectrum $\{s_i\}$, which we do not have); and the derivation assumes a locally quadratic landscape with a static spectrum and diagonal, uncorrelated noise—the same model-level assumptions MPL itself rests on. All real-data experiments are on one public dataset at three scales (the simulation of §4(i′) is a controlled cross-check, not a substitute); broader scales and architectures are the natural next steps.

7 Conclusion

Existing loss-curve laws are adiabatic approximations. We identified, derived from AdamW's preconditioned dynamics, and validated the leading non-adiabatic correction: a rate-dependent term whose relaxation time follows the classical $\tau \propto 1/\eta$ ($p = 1.00 \pm 0.18$, confirmed for the loss-curve residual) and whose scale-invariant shape enables parameter-free cross-scale prediction (-44% on held-out sharp decays). The loss curve depends not only on how much LR has been spent, but on how fast it is being spent.

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